Dilan Karaguler

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Research Interests

Applied algebraic topology and topological data analysis; path homology and Mayer (N-nilpotent) homology of directed graphs and path complexes; persistent homology; topology-informed graph machine learning and geometric deep learning, with applications to molecular, dynamical and engineering data.

Education

Michigan State University — East Lansing, MI, US
Ph.D. in Mathematics (Candidate) — August 2022 – May 2027 (Expected)
- Advisors: Prof. Ekaterina Rapinchuk and Prof. Guowei Wei

Middle East Technical University (METU) — Ankara, Türkiye
M.S. in Mathematics — September 2019 – September 2021
- Thesis: *A Survey on Multidimensional Persistence Theory*
- Advisor: Assoc. Prof. Semra Pamuk

Middle East Technical University (METU) — Ankara, Türkiye
B.S. in Mathematics, Honor Student — June 2014 – June 2019

Publications

D. Karaguler and G.-W. Wei. *Mayer Path Homology.* Foundations of Data Science, accepted (2026). doi:10.3934/fods.2026025, arXiv:2605.16525.

D. Karaguler. *A Survey on Multidimensional Persistence Theory.* M.S. Thesis, Middle East Technical University (2021).

D. Karaguler and G.-W. Wei. *Persistent Mayer Path Laplacians.* In preparation (2026).

Talks

Mayer Path Homology — April 2026
TDA Seminar, Dept. of Computational Mathematics, Science and Engineering, MSU — East Lansing, MI, US

Interaction Topology in TDA — October 2025
Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU — East Lansing, MI, US

Mayer Topology in TDA — October 2025
Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU — East Lansing, MI, US

From Data to Barcodes — April 2021
Graduate Student Seminar, Middle East Technical University — Ankara, Türkiye

Poster Presentations

Mayer Path Topology and Persistence — September 2025
SIAM Great Lakes Section Meeting, Illinois Institute of Technology — Chicago, IL, US

Mayer Path Topology and Persistence — March 2025
Mathematics of Data, Dynamics, and Life Sciences Conference, UC Irvine — Irvine, CA, US

Research Experience

Graduate Researcher — August 2022 – Present
Michigan State University — East Lansing, MI, US
- Develop Mayer path homology, a homology theory for directed graphs built from N-nilpotent differentials on path complexes; published in *Foundations of Data Science* (with G.-W. Wei).
- Implement topological feature extraction (path complexes, Mayer homology and Laplacians, persistent homology) in Python with GUDHI, NumPy and PyTorch, and combine these features with graph neural networks.

Research Mentor — May 2023 – June 2023
Topology RTG, Michigan State University — East Lansing, MI, US
- Mentored an undergraduate researcher on a project in Morse theory: project design, methodology and research presentation; the student went on to major in mathematics.

Industry Experience

Software Engineer Intern May 2026 – August 2026
Intel Corporation Oregon, US

- Built an end-to-end graph transformation pipeline in Python that extracts schematic topology directly from chip layout files, without pre-existing schematic metadata.
- Modeled low-level device instances as graph nodes and extracted a local subgraph around each instance to capture its connectivity.
- Developed graph aggregation and embedding routines producing a vector representation per device, used to classify components as analog, digital or fill devices.
- Worked with Design Yield and Manufacturability engineers to evaluate the prototype for automated structural recognition in layout verification.

Graduate Technical Intern May 2025 – August 2025
Intel Corporation Oregon, US

- Performed spatial pattern analysis on semiconductor chip layer images with KLayout and Python to identify and cluster recurring structures across layout datasets.
- Applied graph neural networks to search for and classify similar patterns across multi-layer designs.
- Partnered with cross-functional yield teams to bring ML-based pattern recognition into automated yield-analysis workflows.

Projects

Persistent Mayer Homology with Dynamic Mode Decomposition April 2026
Python, NumPy, SciPy, GUDHI GitHub

- Built a reproducible pipeline combining Vietoris–Rips filtrations, Mayer homology and Dynamic Mode Decomposition (DMD) to track time-varying topological features.
- Computed Mayer Betti trajectories on synthetic 3D point dynamics; a rank-10 DMD model reconstructed a 500-frame dataset with about 3.6% relative error and supported multi-step forecasting.

Homology-Fed Neural Network Framework November 2024
Python, PyTorch, GUDHI, NumPy GitHub

- Wrote a Python library computing Mayer homology and path-complex Laplacians on graph-structured data.
- Used the resulting topological features for molecular property prediction on the QM9 dataset with graph neural networks.

Cellular Automata on Curved Surfaces March 2024
Python, NumPy, Matplotlib GitHub

- Simulated Conway's Game of Life on a square, torus and sphere with surface-specific neighbor rules, and analyzed how board size and initial density affect population persistence.

Teaching Leadership & Mentoring

Lead Teaching Assistant August 2024 – Present
Department of Mathematics, Michigan State University East Lansing, MI, US

- **2026 – 2027:** Co-lead, with Eloy Moreno-Nadales, a yearlong workshop series that prepares first-year graduate students for their first teaching assignments. Sessions so far: an introduction run as a gallery walk, with a "Math PhD passport" and department scavenger hunt connecting new students with senior graduate students; and *Three Aspects of Teaching* (clarity and delivery, communication and language, engagement and cohort needs), built on the TAs' own classroom observations to prepare their mock teaching.
- **Fall 2025 (MTH 299):** Prepared recitation worksheet materials, gathered TA feedback for the instructors, and substituted for recitation leaders when needed.
- **Spring 2025 (MTH 133):** Ran TA meetings, prepared weekly recitation worksheets, and set up Teams tools to track the TA timeline and grading deadlines.
- **Fall 2024:** Ran the orientation for undergraduate learning assistants (ULAs) starting in August, held mock-teaching and feedback sessions, and observed ULAs and GTAs during the semester to give feedback; as MTH 133 pod leader, ran meetings with recitation leaders and prepared weekly recitation worksheets.

Graduate Lead, Center for Instructional Mentoring (CIM) Summer 2024
Department of Mathematics, Michigan State University East Lansing, MI, US

- Mentored first-time graduate instructors through weekly check-ins, one-on-one conversations and non-evaluative teaching feedback, and introduced the teaching technology they could use.
- Created a teaching handbook built on interviews with senior GTAs about the courses they taught, describing each course's expectations so junior graduate students can choose assignments that fit their needs.

Mentor, Academic Culture and English Skills (ACES) Program Summer 2024
Michigan State University East Lansing, MI, US
- Mentored incoming international graduate students preparing to become MSU teaching assistants, introducing U.S. classroom culture, university structure and cross-cultural communication, and organized social events.

Supervisor, Math Learning Center (MLC) Fall 2022 – Spring 2023, Fall 2024
Michigan State University (Teaching Exemption Fellowship, 2022 – 2023) East Lansing, MI, US
- Managed how tutors were distributed among students and arranged the room to match demand.

Student feedback: dilankaraguler.github.io/teaching/what-students-say

Courses Taught

MTH 314 — Matrix Algebra with Computational Applications Fall 2026
Instructor (1 section), Michigan State University East Lansing, MI, US
- Prepare and deliver lectures, facilitate group work, and hold help-room hours.

MTH 299 — Transitions Fall 2025
Teaching Assistant (2 sections), Michigan State University East Lansing, MI, US
- Co-taught with Ester Sgallova: led weekly recitations, wrote homework solutions, graded homework and quizzes, and held help-room hours.

MTH 314 — Matrix Algebra with Computational Applications Spring 2025
Teaching Assistant (1 section), Michigan State University East Lansing, MI, US
- Graded homework and the final exam, ran quiz and exam review sessions, and held help-room hours.

MTH 133 — Calculus II Fall 2024
Teaching Assistant (1 section) and pod leader, Michigan State University East Lansing, MI, US
- Led a recitation section; as pod leader, ran meetings with the recitation leaders and prepared the weekly recitation worksheets.

MTH 235 — Differential Equations Spring 2024
Teaching Assistant (3 sections), Michigan State University East Lansing, MI, US
- Wrote recitation notes with lecture summaries and practice problems, graded quizzes, held help-room hours, and followed up with students in online sections.

MTH 133 — Calculus II Fall 2023
Teaching Assistant (3 sections: 2 online, 1 in person), Michigan State University East Lansing, MI, US
- Prepared weekly lecture summaries for the recall session before group work and worksheets; graded and proctored worksheets and quizzes.
- Held help-room hours and tracked student progress, reaching out to students as needed, especially in online sections.

MTH 133 — Calculus II Summer 2023
Teaching Assistant (1 online asynchronous section), Michigan State University East Lansing, MI, US
- Contacted students weekly by email and tracked their course activity.

MTH 154, 153, 120, 119 — Calculus I and II January 2020 – June 2022
Graduate Teaching Assistant, Middle East Technical University Ankara, Türkiye
- MTH 154 Calculus II for Mathematics Students (Spring 2022), MTH 153 Calculus I for Mathematics Students (Fall 2021), MTH 120 Calculus II for Engineering Students (Spring 2021), MTH 119 Calculus I for Engineering Students (Fall 2020).
- Led recitations and office hours, graded, and proctored online and in-person exams.

Calculus for Mathematics Students March 2018 – June 2019
Undergraduate Teaching Assistant, Middle East Technical University Ankara, Türkiye
- Graded homework and held review hours.

Awards, Fellowships & Scholarships

Doctoral Student Summer Internship Fellowship, The Graduate School, Michigan State University — Summer 2026

Excellence in Teaching Award, Department of Mathematics, Michigan State University — May 2025

Colleges Online Learning Academy (COLA) Fellowship, Michigan State University — Summer 2024

Teaching Exemption Fellowship, Michigan State University — 2022 – 2023

RONESANS Scholarship, RONESANS — 2018 – 2019

UNIGENC Scholarship, UNIGENC Association — 2015 – 2019

Merit Scholarship, Turkish Ministry of National Education — 2010 – 2014

Service, Leadership & Outreach

Graduate Student Representative, Graduate Studies Committee 2025 – 2026
Department of Mathematics, Michigan State University East Lansing, MI, US

- Represented all mathematics graduate students at committee meetings with faculty.
- Planned and prepared Responsible Conduct of Research (RCR) workshops with the Graduate Chair.
- Met with prospective graduate students and hosted graduate student town hall meetings on the state of the department.
- Served as a point of contact for graduate students with grievances and directed them to department and university resources.

President / Board Member August 2022 – August 2025
Association for Women in Mathematics (AWM), MSU Chapter East Lansing, MI, US

- Organized seminars, career workshops, informational sessions on academic life and social events for women in mathematics.

Member, First-Generation/Low-Income (FGLI) Group September 2024 – May 2025
Department of Mathematics, Michigan State University East Lansing, MI, US

- Took part in community gatherings and social events with first-generation and low-income undergraduates.

Organizing Committee Member April 2024
Graduate Student Topology and Geometry Conference, Michigan State University East Lansing, MI, US

- Handled communication with speakers and attendees, room assignments and talk scheduling, and moderated sessions.

President February 2015 – December 2018
METU Math Club Ankara, Türkiye

- Organized seminars, workshops and regular reading groups in topology, algebra and other areas.
- Led Math Club reading sessions with first-year students in abstract algebra and number theory, preparing each student to lead a discussion of part of the material.

Technical Skills

Programming: Python, R, SQL
Methods: Graph Neural Networks, Graph Transformers, Topological Data Analysis, Persistent Homology, Vietoris–Rips Filtrations, Dynamic Mode Decomposition, Supervised/Unsupervised Learning, Model Evaluation
Libraries: PyTorch, PyTorch Geometric, NetworkX, GUDHI, NumPy, SciPy, Pandas, Scikit-Learn, TensorFlow, Keras, Matplotlib, XGBoost, LightGBM
Tools: Git, Linux/Unix, DuckDB, SQLite, Jupyter, KLayout, LaTeX
Domain Knowledge: Design for Manufacturability (DFM), Semiconductor Yield Analysis, Schematic Extraction, Chip Layout Analysis

References

Available upon request.